

Home > Stainless Steel 304 - 1.4301 Data ...

Stainless Steel 304 1.4301

Stainless steel 304 and stainless steel 304L are also known as 1.4301 and 1.4307 respectively. Type 304 is the most versatile and widely used stainless steel. It is still sometimes referred to by its old name 18/8 which is derived from the [nominal composition](#) of type 304 being 18% chromium and 8% nickel. Type 304 stainless steel is an austenitic grade that can be severely deep drawn. This property has resulted in 304 being the dominant grade used in applications like sinks and saucepans. Type 304L is the low carbon version of 304. It is used in heavy gauge components for improved weldability. Some products such as plates and pipes may be available as “dual certified” material that meets the criteria for both 304 and 304L. 304H, a high carbon content variant, is also available for use at high temperatures. Properties given in this data sheet are typical for flat-rolled products covered by ASTM A240/A240M. It is reasonable to expect specifications in these standards to be similar but not necessarily identical to those given in this data sheet.

Application

Saucepans
Springs, screws, nuts & bolts
Sinks & splash backs
Architectural panelling
Tubing
Brewery, food, dairy and pharmaceutical production equipment
Sanitary ware and troughs

Supplied Forms

Sheet
Strip
Bar
Plate
Pipe
Tube
Coil
Fittings

Alloy Designations

Stainless steel grade 1.4301/304 also corresponds to: S30400, 304S15, 304S16, 304S31 and EN58E.

Corrosion Resistance

304 has excellent corrosion resistance in many environments and when in contact with different corrosive media. Pitting and crevice corrosion can occur in environments containing chlorides. Stress corrosion cracking can occur above 60°C.

Heat Resistance

304 has good resistance to oxidation in intermittent service up to 870°C and in continuous service to 925°C. However, continuous use at 425- 860°C is not recommended. In this instance 304L is recommended due to its resistance to carbide precipitation. Where high strength is required at temperatures above 500°C and up to 800°C grade 304H is recommended. This material will retain aqueous corrosion resistance.

Fabrication

Fabrication of all stainless steels should be done only with tools dedicated to stainless steel materials. Tooling and work surfaces must be thoroughly cleaned before use. These precautions are necessary to avoid cross contamination of stainless steel by easily corroded metals that may discolour the surface of the fabricated product.

Cold Working

304 stainless steel readily work hardens. Fabrication methods involving cold working may require an intermediate annealing stage to alleviate work hardening and avoid tearing or cracking. At the completion of fabrication a full annealing operation should be employed to reduce internal stresses and optimise corrosion resistance.

Hot Working

Fabrication methods like forging, that involving hot working should occur after uniform heating to 1149-1260°C. The fabricated components should then be rapidly cooled to ensure maximum corrosion resistance.

Machinability

304 has good machinability. Machining can be enhanced by using the following rules: Cutting edges must be kept sharp. Dull edges cause excess work hardening. Cuts should be light but deep enough to prevent work hardening by riding on the surface of the material. Chip breakers should be employed to assist in ensuring swarf remains clear of the work. Low thermal conductivity of austenitic alloys results in heat concentrating at the cutting edges. This means coolants and lubricants are necessary and must be used in large quantities.

Heat Treatment

304 stainless steel cannot be hardened by heat treatment. Solution treatment or annealing can be done by rapid cooling after heating to 1010- 1120°C.

Weldability

Fusion welding performance for type 304 stainless steel is excellent both with and without fillers. Recommended filler rods and electrodes for stainless steel 304 is grade 308 stainless steel. For 304L the recommended filler is 308L. Heavy welded sections may require post-weld annealing. This step is not required for 304L. Grade 321 may be used if post-weld heat treatment is not possible.

Chemical Compositions^{a)}

Element	% Present
Carbon (C)	0.07
Chromium (Cr)	17.50 - 19.50
Manganese (Mn)	2.00
Silicon (Si)	1.00
Phosphorous (P)	0.045
Sulphur (S)	0.015 ^{b)}
Nickel (Ni)	8.00 - 10.50
Nitrogen (N)	0.10
Iron (Fe)	Balance

Mechanical properties

Property	Value
Comprehensive Strength	210 MPa
Proof Stress	210 Min MPa
Tensile Strength	520 - 720 MPa
Elongation	45 Min%

Physical Properties

Property	Value
Density	8,000 Kg/m ³
Melting Point	1450 °C
Thermal Expansion	17.2 x 10 ⁻⁶ /K
Modulus of Elasticity	193 GPa
Thermal Conductivity	16.2W/m.K
Electrical Resistivity	0.072 x 10 ⁻⁶ Ω .m

Editor

thyssenkrupp Materials (UK) LtdCox's LaneCradley HeathWest MidlandsB64 5QU

Important Note

Information given in this data sheet about the condition or usability of materials respectively products are no warranty for their properties, but act as a description. The information, we give on for advice, comply to the experiences of the manufacturer as well as our own. We cannot give warranty for the results of processing and application of the products.

Quicklinks

[Stainless Steel Home Page](#)

[Stainless Steel Datasheets](#)

Other datasheets

In addition to the above, we also have datasheets that detail the mechanical properties and chemical composition of the following stainless steel grades:

301 - 14310

304L - 14307

310 - 14845

316 - 14401

316Ti - 14571

321H - 14878

409 - 14512

441 - 14509

14435

14724

303 - 14305

309 - 14828

314 - 14841

316L - 14404

321 - 14541

347 - 14550

430 - 14016

904L - 14539

14713